

Shift Register Display

Send one bit at a time; reveal one stable digit

The displayed glyph and three named test points explain the transfer without Serial.

Lesson	010 — component	Time	120–180 minutes
Prerequisites	Lessons 001–009	Board	Arduino Mega 2560
Evidence	glyph, TP-DATA, TP-CLOCK, TP-LATCH	Status	host verified; bench open
Safety	E1: USB-powered logic and resistor-limited LED segments only		

1 Mission and evidence chain

Build a decimal counter from two high-level objects: `ShiftRegisterOutput` owns the three control pins, while `SevenSegmentDisplay` turns a glyph into the register’s eight output bits. The sketch reads in the same order as the circuit:

time → choose digit → serialize bits → latch once → stable glyph.

The D13 diagnostic LED’s 250 ms pulse proves that both components acquired their pins. A 750 ms dark separator follows before counting begins. The display proves the current glyph. TP-DATA, TP-CLOCK, and TP-LATCH expose the electrical transfer to a logic probe or oscilloscope. These observations prove different claims; none alone proves the entire chain.

1.1 Learning outcomes

You will be able to:

- distinguish a shift clock from the output latch;
- predict the register byte for a decimal glyph;
- explain why the visible display remains stable while new bits shift;
- use deterministic RAII ownership and a nonblocking update loop;
- separate acquisition, behavior, and safe-state evidence.

Safety checkpoint. The software target is a TI SN74HC595N in PDIP-16. The Elegoo inventory identifies the display only by family, not by marking or pinout. Photograph and record the actual display marking, then establish its package pins from a primary datasheet or an unpowered, resistor-limited continuity map before connecting any Q output. Seven-segment package pinouts vary. A common-anode display wired as common-cathode gives misleading results and can exceed an LED rating. Remove USB power before moving wires or meter clips.

2 Parts and current budget

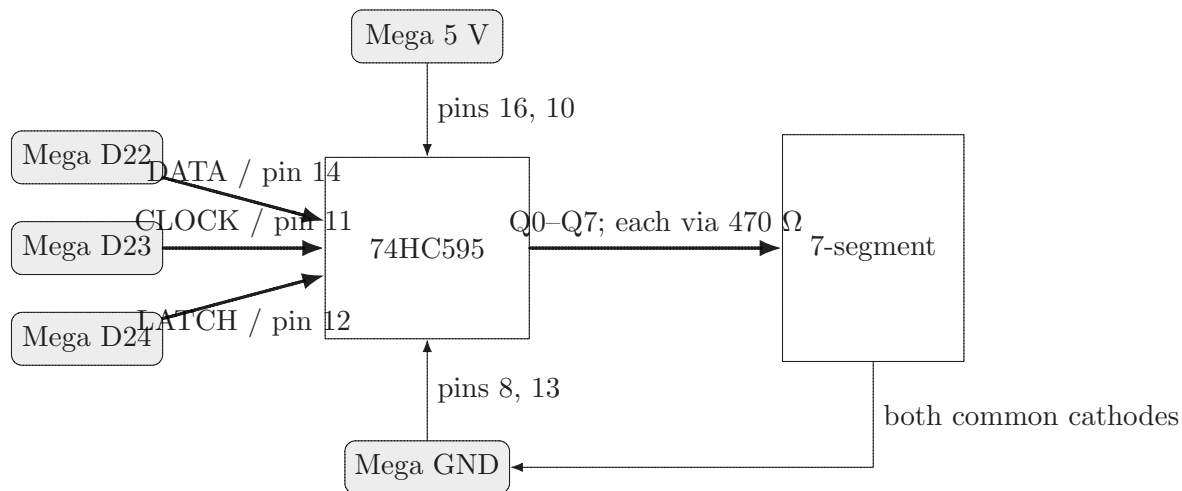
Quantity	Part	Requirement
1	Arduino Mega 2560	USB-powered reference board
1	Shift register	TI SN74HC595N, PDIP-16, marking and lot code recorded
1	Seven-segment display	One digit, common cathode, exact marking and pinout recorded
8	470 Ω resistors	One per segment, including decimal point
1	100 nF ceramic capacitor	Register supply bypass at pins 16 and 8
1	Breadboard and jumpers	Short, inspectable connections
1	Analyzer or oscilloscope	Required: at least three digital channels and 1 MHz sampling
1	USB cable and supply	Exact cable and supply identities recorded

For each lit segment, calculate

$$I_{\text{segment}} = \frac{V_{\text{CC}} - V_F}{470 \, \Omega}.$$

Record the display datasheet's forward-voltage range and absolute maximum. Measure the voltage across each active 470 Ω resistor and calculate $I = V_R/R_{\text{measured}}$; do not infer current from brightness. The lesson limit is 8 mA per segment and 40 mA total display current; increase the resistor value if either calculation exceeds its limit. These conservative lesson limits do not replace the register and display datasheets. Also record USB supply identity, unloaded and loaded 5 V rail, and total USB current with glyph 8. Stop if the rail leaves 4.75–5.25 V or total USB current exceeds 100 mA.

3 Authoritative wiring

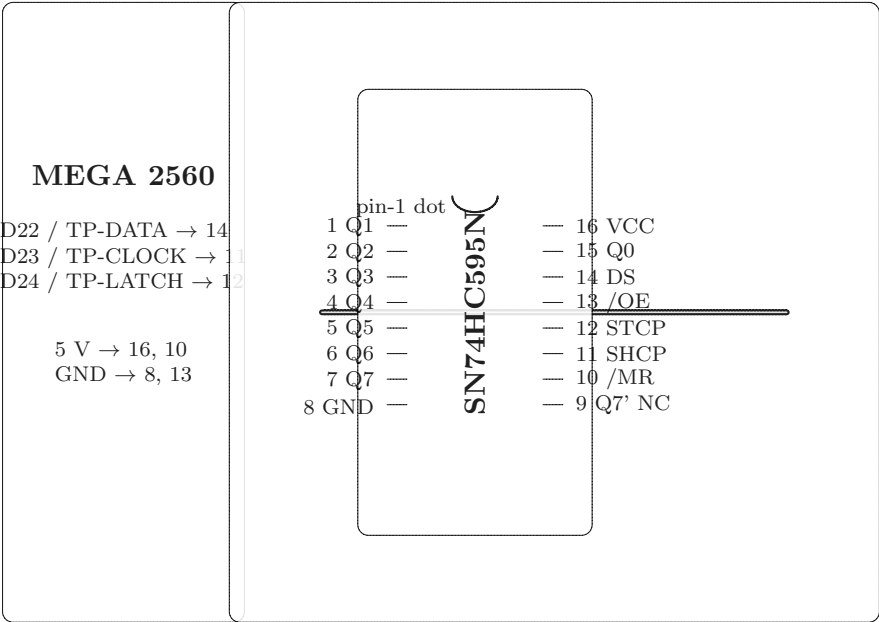


Pin-by-pin connection list

- D22 / TP-DATA → DS, 74HC595 pin 14.
- D23 / TP-CLOCK → SHCP, pin 11.
- D24 / TP-LATCH → STCP, pin 12.
- 5 V → VCC pin 16 and active-low clear pin 10.
- GND → GND pin 8 and active-low output-enable pin 13.
- Q0 (pin 15), Q1–Q6 (pins 1–6) → segments a through g; Q7 (pin 7) → decimal point. Every path has its own 470 Ω resistor.
- Both qualified display common-cathode pins → GND, using the package-pin numbers established on Build Plate B.

Place a 100 nF ceramic bypass capacitor directly between pins 16 and 8. Orient the PDIP notch upward and count pins counterclockwise: pin 1 is upper left, pin 8 lower left, pin 9 lower right, and pin 16 upper right. Straddle the breadboard center trench so opposite IC pins never share a contact strip. The pencil plate shows every register connection and resistor path, but the display outline is deliberately marking-neutral: write its verified segment terminal numbers into the eight boxes before placement. Do not power a display whose marking and common-cathode pinout are unresolved. In this introductory three-wire circuit, active-low OE is tied to GND, so the output drivers are enabled before the sketch runs. The 74HC595 storage register's power-up state is not specified; reset or power-up can therefore show an arbitrary, possibly bright, segment pattern until the first blank or glyph is latched. The software lifecycle cannot prevent that startup transient.

A later hardware upgrade gives OE its own Mega output and a pull-up resistor. The pull-up keeps the register outputs disabled while the Mega resets; software then shifts and latches the blank value before driving OE low. That fourth control signal is required when blank output during power-up is part of the circuit contract.



Build Plate A is an orientation plate, not a netlist: it fixes the notch, pin-1 dot, counterclockwise numbering, and placement across the breadboard trench. The pin-by-pin list above is authoritative for every control, supply, clear, enable, test-point, and bypass net. Place the 100 nF capacitor physically beside the IC between pins 16 and 8.

Build Plate B — unpowered display specimen map

Marking/photo _____ Primary source _____

Common type _____ Cathode package pins _____,

Logical path	Q0/a	Q1/b	Q2/c	Q3/d	Q4/e	Q5/f	Q6/g	Q7/dp
595 pin	15	1	2	3	4	5	6	7
470 Ω	☐	☐	☐	☐	☐	☐	☐	☐
Display pin	—	—	—	—	—	—	—	—

Both qualified common-cathode pins go to GND. Until every Plate B blank is resolved from the actual specimen, stop: the powered circuit is not authorized.

4 Interface and ownership

```
constexpr adk::ShiftRegisterPins displayPins = {22, 23, 24};

adk::Runtime          runtime;
adk::SevenSegmentDisplay display (
    runtime.resources (),
    displayPins,
    adk::SevenSegmentPolarity::CommonCathode);

display.initialize ();
display.show (adk::SevenSegmentGlyph::Five);
display.blank ();
display.shutdown ();
```

`SevenSegmentDisplay` composes one `ShiftRegisterOutput`. Initialization claims D22, D23, and D24 transactionally. A conflict leaves no partial claim. Repeated initialization succeeds without another transfer. `show()` accepts digits, hexadecimal glyphs, dash, blank, and an optional decimal point.

For common cathode, a one bit lights its segment. For common anode the API inverts all eight output bits. The wiring and constructor polarity must agree. The supported mapping is Q0–Q6 to a–g and Q7 to the decimal point.

`shutdown()` first shifts the polarity-correct blank value, then makes DATA, CLOCK, and LATCH high impedance and releases their claims. Destruction does the same during normal scope exit or exception unwinding in a caller. ADK itself neither throws nor allocates.

4.1 A transfer without update flicker

For each byte, `ShiftRegisterOutput::show()` holds LATCH low, presents the most-significant remaining bit on DATA, and pulses CLOCK. After eight clock pulses it pulses LATCH once. The output register changes as one visible step, so intermediate shifted bytes are not shown during normal updates. This does not control the register’s unspecified state before the first software latch while OE remains tied low.

Predict: glyph 8 uses segments a–g, so its common-cathode value is 0x7f. **Observe:** seven segments lit and decimal point dark. **Interpret:** the stable output matches the glyph encoding. It does not prove clock timing margins; TP-CLOCK and TP-LATCH supply that evidence.

5 Narrative run loop

The canonical sketch declares the runtime, display, and diagnostic LED in dependency order. `setup()` reads “acquire circuit, pulse ready.” `loop()` observes the acquisition, separator, count, and shutdown deadlines before choosing and showing a digit. At 120 seconds it blanks the display and D13, holds that inactive evidence for 250 ms, releases D22–D24 then D13, and halts. Reset starts the same bounded sequence again. No `delay()` blocks observation.

```
void loop ()
{
    if (!ready)
    {
        return;
    }
}
```

```
    const adk::TimePoint now (millis ());

    if (!countIsDue (now))
    {
        return;
    }

    const adk::SevenSegmentGlyph digit = chooseDigit ();
    showDigit (digit);
}
```

The complete, compile-authoritative sketch is the Lesson 010 sketch download. Build and upload it only through repository targets; no particular IDE is required.

6 CLI experiment

```
make bootstrap
arduino-cli core list
git rev-parse HEAD
make arduino-Lesson010ShiftRegisterDisplay
make upload-Lesson010ShiftRegisterDisplay PORT=/dev/ttyACM0
make observe-new LESSON=010
make observe-check CARD=hardware/acceptance/lesson-010.md
```

Serial is optional supporting evidence:

```
make monitor PORT=/dev/ttyACM0 BAUD=115200
make serial-log PORT=/dev/ttyACM0 BAUD=115200
```

The canonical sketch does not require or emit Serial messages. A missing serial log therefore says nothing about display correctness.

6.1 Predict, observe, interpret

Place	Prediction	Interpretation
D13	250 ms on, then a 750 ms dark separator	All four output pins were acquired; not proof of segment wiring
Display	Zero during the evidence prelude, then one increment each second	Glyph selection and latched outputs agree visibly
TP-DATA	Eight sampled levels per update, most significant bit first	Decode and compare the byte with the shown glyph
TP-CLOCK	Exactly eight pulses per update	Eight bits were shifted
TP-LATCH	One pulse after the eight clocks	The visible output changes atomically
D22–D24 after shutdown	Each preserves a 10 k Ω /10 k Ω midpoint near half rail	Endpoint safe state; test one pin at a time

6.2 Experiment sequence

1. With USB removed, inspect every IC pin, resistor, display common, and the bypass capacitor against the exact part datasheets.
2. Predict the first three glyphs and the CLOCK:LATCH pulse count.
3. Attach probes with power removed. Reference every measurement to Mega GND. Apply USB power only after a second inspection.
4. Confirm the 250 ms D13 pulse and 750 ms dark separator, then the visible 0–9 sequence.
5. A three-or-more-channel, 1 MHz-or-faster analyzer or oscilloscope is required. With USB removed, connect CH0 to TP-DATA/D22, CH1 to TP-CLOCK/D23, CH2 to TP-LATCH/D24, and analyzer ground to Mega GND. Inspect the clips, repower, use a rising-edge TP-LATCH trigger, and save a capture with the preceding eight clocks. Decode TP-DATA MSB first and compare that byte with the displayed glyph.
6. At the fixed 120-second deadline, confirm that the display and D13 are inactive for 250 ms before the pins release. Remove USB. Install the 10 k Ω /10 k Ω midpoint fixture on D22 only, inspect it, repower, and record the midpoint; then remove USB before removing the fixture. Repeat that complete remove/install/inspect/repower/measure/remove sequence separately for D23, D24, and D13. Never move the fixture while powered. A high-impedance pin leaves the midpoint near half the measured rail.

7. Press reset and record the OE-low transient duration from reset edge to the first software latch. Confirm the pulse, separator, zero, and count sequence repeat. Remove USB and verify all parts are dark and cool before moving probes.

Host-verified is the current release status. Do not fill the measured columns or mark hardware accepted until the actual Mega circuit has been observed and recorded.

7 Diagnosis by evidence

Observation	Distinguishes	Next check
D13 dark	acquisition failed or setup stopped	pin conflicts, exact status in host test, then D13 circuit
D13 on; display dark	control acquired but display path unknown	5 V/GND, OE low, clear high, display polarity
Wrong stable glyph	output path exists; mapping differs	Q0–Q7 order and segment a–dp datasheet pins
Partial glyph	one or more segment paths absent	resistor, LED segment, Q output continuity unpowered
Flicker or mixed glyph	latch behavior or supply integrity suspect	TP-LATCH pulse count and 100 nF bypass
No CLOCK pulses	transfer did not occur	TP-DATA/TP-CLOCK, initialization, pin conflict
Eight clocks; no latch	shift register changed internally only	TP-LATCH wiring and pin 12
Chip or resistor hot	unsafe current or wiring	remove USB immediately; recheck identity and current budget

7.1 Fault injections without rewiring live

- In a host test, pre-claim CLOCK. Expect `ResourceBusy`, no partial DATA/LATCH ownership, and no output transfer.
- Construct an object with duplicate data and clock pins. Expect `InvalidArgument` before a claim or hardware write.
- Request every glyph twice and compare the recording fake's traces. They must match byte for byte.
- Advance timestamps across 32-bit wrap in the example timing model. The elapsed-time decision must remain correct.

8 Design exercises

1. Write the common-cathode bytes for 0, 1, 2, and dash. Then invert each byte for common anode.
2. Why is one resistor on the display common pin not equivalent to one resistor per segment?
3. Extend the counter to hexadecimal without changing `ShiftRegisterOutput`.
4. Add a decimal-point heartbeat that does not block the counter. Name the independent state and deadline.
5. Sketch two cascaded 74HC595 registers. Which signal is shared, which output feeds the next DATA input, and which resource remains owned?
6. Explain what a stable digit proves that a TP-CLOCK capture does not, and vice versa.

9 Bench acceptance record

Date	_____	Reviewer	_____
Sketch commit	_____	Sketch path	_____
Firmware artifact/hash	_____	Source tree state	_____
CLI version	_____	AVR core version	_____
Board	Mega 2560 revision: _____	USB cable	_____
USB supply/model	_____	Supply current limit	_____
74HC595 marking	_____	Display marking	_____
Display polarity	_____	Resistors	_____
Unloaded 5 V	_____	Loaded 5 V, glyph 8	_____
USB current, glyph 8	_____	Display-current total	_____

9.1 Glyph-8 resistor and current worksheet

Path	a/Q0	b/Q1	c/Q2	d/Q3	e/Q4	f/Q5	g/Q6	dp/Q7
R, Ω	_____	_____	_____	_____	_____	_____	_____	_____
V_R, V	_____	_____	_____	_____	_____	_____	_____	dark
$I = V_R/R, \text{mA}$	_____	_____	_____	_____	_____	_____	_____	0

Glyph 8 lights a–g and leaves dp dark. Sum the seven calculated currents and compare that value with the 40 mA display limit; separately record total USB current, which also includes the board and register. Attach a signed worksheet if the available instrument log provides these same eight path fields.

9.2 Required observations

Check	Pass	Recorded evidence
Unpowered wiring matches both datasheets	☐	
D13 gives a 250 ms acquisition pulse and 750 ms separator	☐	
Digits 0–9 are stable and correctly mapped	☐	
Analyzer record names channels, sample rate, trigger, and decoded TP-DATA byte	☐	
TP-CLOCK shows eight pulses before one TP-LATCH edge	☐	
All eight resistor values/drops/currents and glyph-8 total are recorded	☐	
Loaded/unloaded rail and glyph-8 total USB current meet limits	☐	
Startup transient with OE tied low is observed and recorded	☐	
First software latch establishes the documented zero	☐	
Reset returns to the documented initial sequence	☐	
Each unpowered-installed midpoint fixture passes after 120 s shutdown	☐	
Power removal leaves every part cool and dark	☐	

Instrument and capture identifiers: _____

Deviations or open questions:

Stop conditions: unexpected heat, unstable USB, a display current above the recorded budget, signal outside the rails, unexpected simultaneous segments, or disagreement between the actual part datasheet and this lesson. Remove USB power physically; software shutdown is not the emergency stop.

Deferred upgrade: repeat the startup and reset observations after adding a pulled-up, software-owned OE signal. Record that outputs remain disabled until a blank byte has been shifted and latched; do not transfer that claim to the three-wire circuit above.

10 What comes next

Lesson 011 keeps this display visible while a deterministic, nonblocking traffic state advances and records pedestrian requests. Lesson 012 composes the display, buttons, and mutually exclusive signal lamps into the next three-lesson project.